

Release Notes of QCS8550 Ubuntu 1.0 for RDK/Grizzly

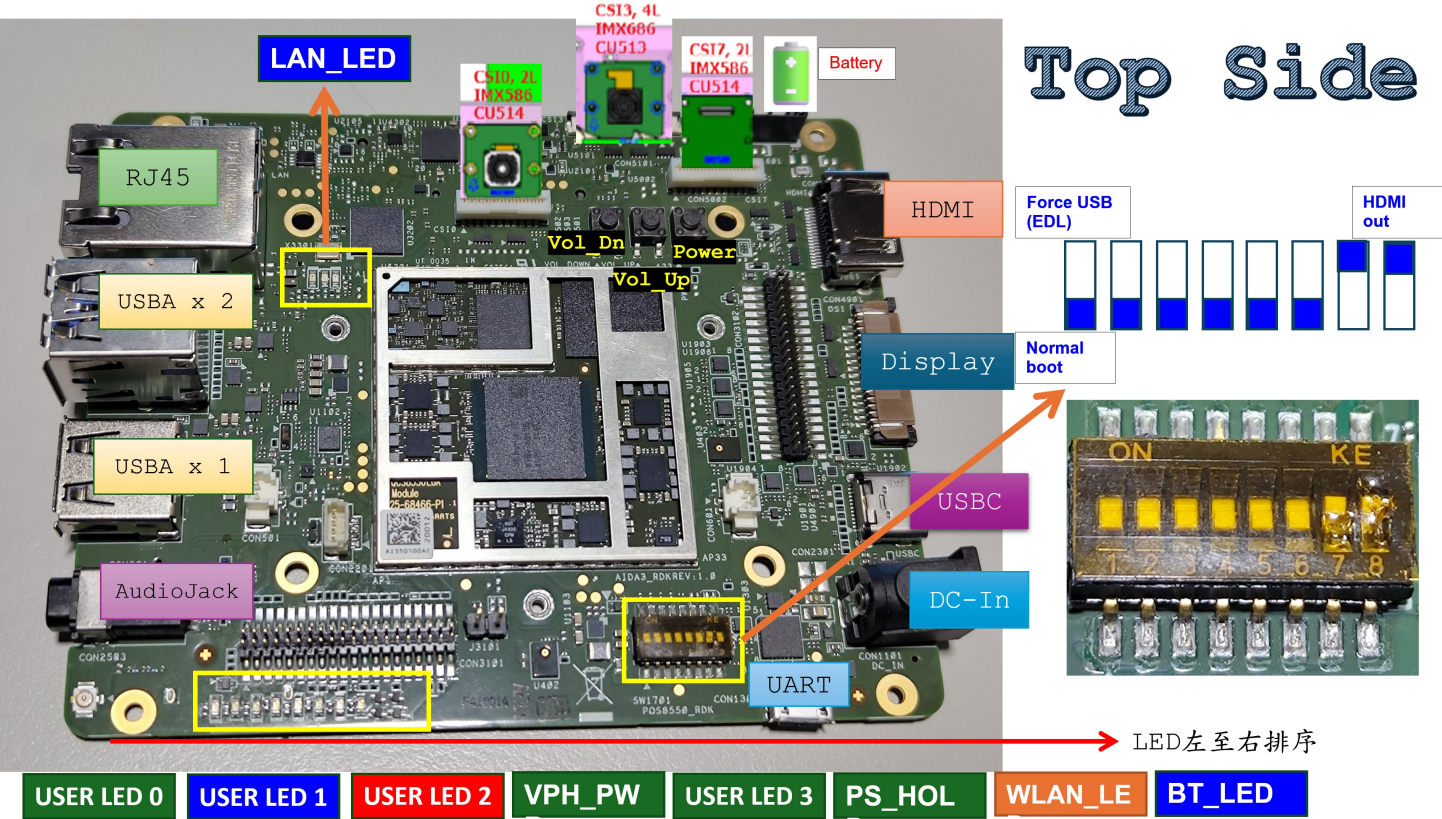
This document describes how to flash, connect to, and validate the QCS8550 Ubuntu 1.0 release on RDK and Grizzly hardware. It targets developers and validation engineers integrating this platform. Follow the steps below to set up connectivity, flash images, configure display and cameras, and run basic multimedia and AI tests.

Release version

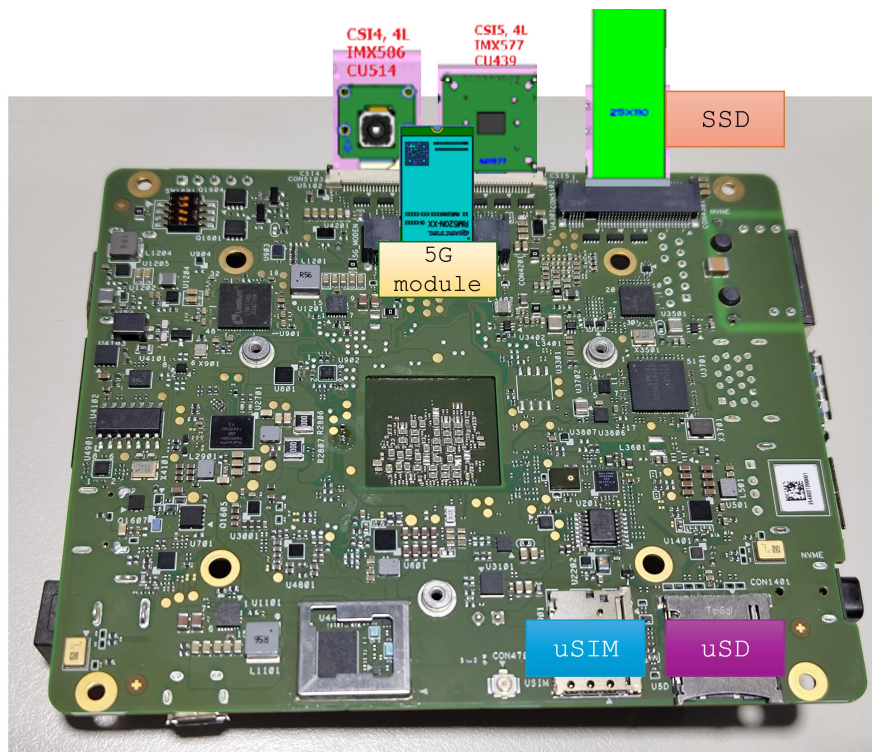
```
adb wait-for-device root
adb wait-for-device shell "cat firmware/verinfo/ver_info.txt"
{
  "Image_Build_IDs": {
    "adsp": "ADSP.HT.5.8-01390-KAILUA_IOT-2",
    "aop": "AOP.H0.5.0-00794-KAILUA_E-1",
    "apps": "LE.PRODUCT.2.3.r1-08400-KAILUA.0-1",
    "apps_kernel": "KERNEL.PLATFORM.2.0.r27-02200-kernel.0-1",
    "apps_le_framework": "LE.FRAMEWORK.2.0.r5-02100-KAILUA.0-1",
    "boot": "BOOT.MXF.2.1.1.c1-00019-KAILUA-1",
    "btfm_hmt": "BTFW.HAMILTON.2.0.1-00444-PATCHZ-2",
    "cdsp": "CDSP.HT.2.8-00963-KAILUA-1",
    "common": "QCS8550.UBUN.1.0-00015-STD.PROD-1",
    "cpucp": "CPUCP.FW.1.0-00085-KAILUA_EXT-1",
    "cpuss_vm": "CPUSS.CPUSYS.VM.1.0-00019-KAILUA_EXT-2",
    "glue": "GLUE.KAILUA_LU.1.0-00045-NOOP_TEST_KAILUA-1",
    "modem": "MPSS.DE.3.0.c3-00044-KAILUA_GEN_PACK-2",
    "tz": "TZ.XF.5.24-00357-ADJKAILUAAAANAZT-1",
    "tz_apps": "TZ.APPS.1.24-00120-KAILUAAAAAANAZT-1",
    "wlan_hmt": "WLAN.IOE_HMT.1.0.2-00453-QCAHMTSWPL_V1.0_V2.0_SILICONZ-1",
    "wlan_wbe": "WLAN.IOE_WKK.1.0-00293-QCAHKSPL_SILICONZ_IOE-1"
  },
  "Metabuild_Info": {
    "Meta_Build_ID": "QCS8550.UBUN.1.0-00015-STD.PROD-1",
    "Product_Flavor": "asic",
    "Time_Stamp": "2026-04-20 09:27:38"
  },
  "Pegabuild_Info": {
    "Pega_Build_Branch": "main.15.1",
    "Pega_Build_Commit": "576728c775dfa24956eb041f8ea23242f468a6aa"
  },
  "Version": "1.0"
}
```

RDK IO ports layout

- Top side



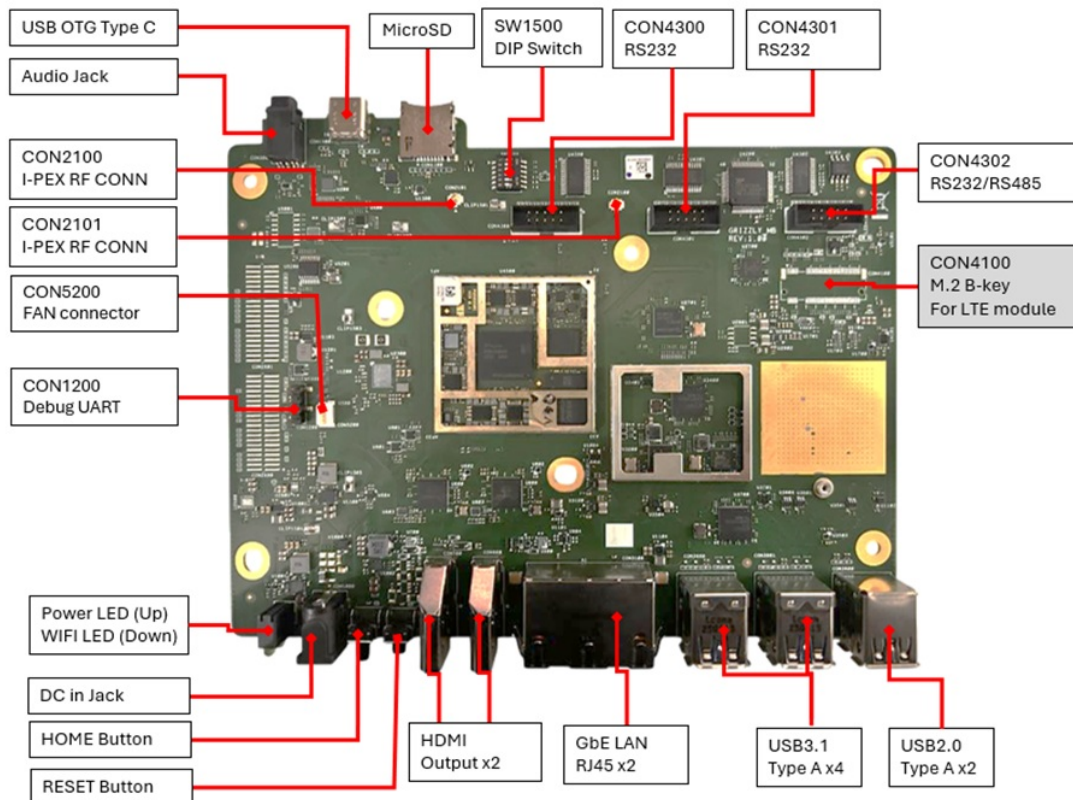
- Bottom side



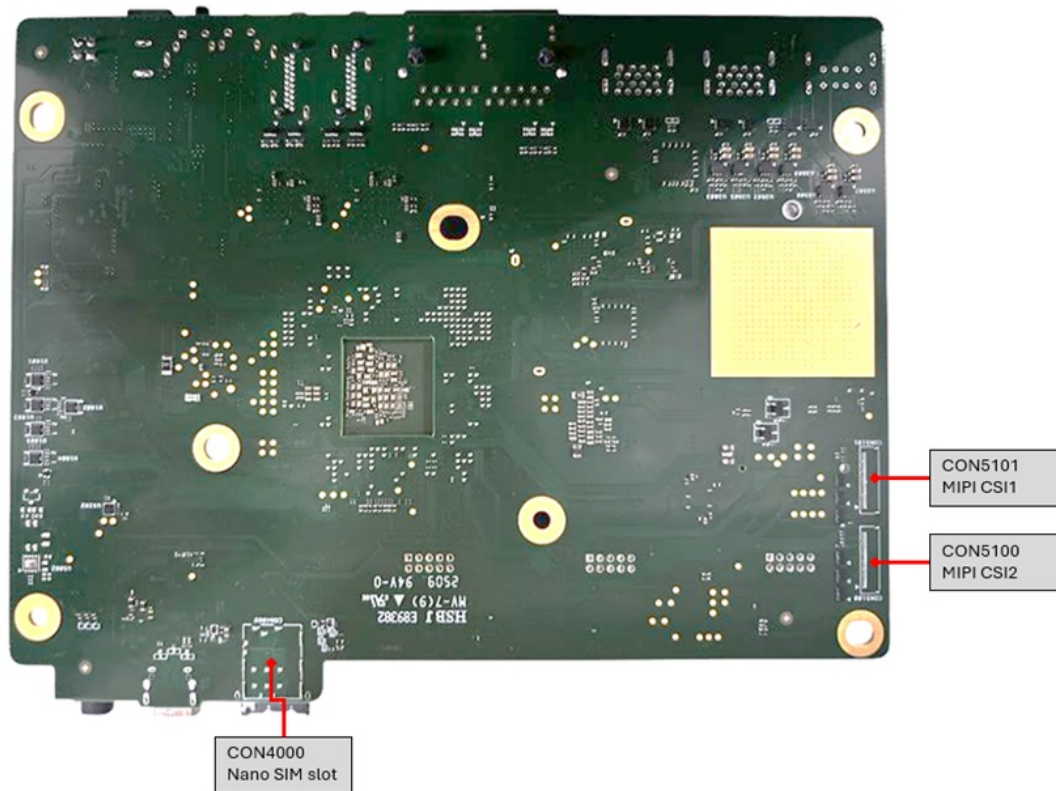
BOT side

Grizzly IO ports layout

- Top side



- Bottom side



Connect through ADB via USBC

To connect the device to the host machine through ADB, follow these steps. - RDK USBC (CON2301) - Grizzly USBC (CON1900)
 1. Install the ADB tool on your PC. 2. Connect the 12V power adaptor to the device first and then USB Type-C between the device and the PC.

Connect through UART for debugging

- RDK USBC (CON1301)
- Grizzly pin header (CON1200)

To connect the device to the host machine through UART, follow these steps. 1. The host machine connects to - RDK UART via a USB2.0 (USBA-MicroUSB) cable. - Grizzly UART via a 4-wired cable. 2. Power on the device. 3. Check the device serial communication (COM) found in the device manager section of the system. 4. Log in to the serial COM tool, such as PuTTY. - Baud rate (typically 115200) - Data bits (8) - Parity (None) - Stop bits (1) - Flow control (None) 5. View the serial log dump on your console. 6. Log in with your account and password.

- Username: root
- Password: oelinux123

Connect through UART (GPIO50, 51) for flight control on RDK

To connect the device to the flight control system through UART, follow these steps. 1. Connect the QCS8550 device to the flight control system via 3 wires (TX, RX and GND). 2. Power on the device. 3. The tty device of QCS8550 is /dev/ttyHS4.



GND

SM_GPIO_50 (Tx)

SM_GPIO_51 (Rx)

Flash images to the device

Note: These methods are only for flashing flatten images. They do NOT support fastboot images.

Method 1: Windows (QsaharaServer.exe and fh_loader.exe)

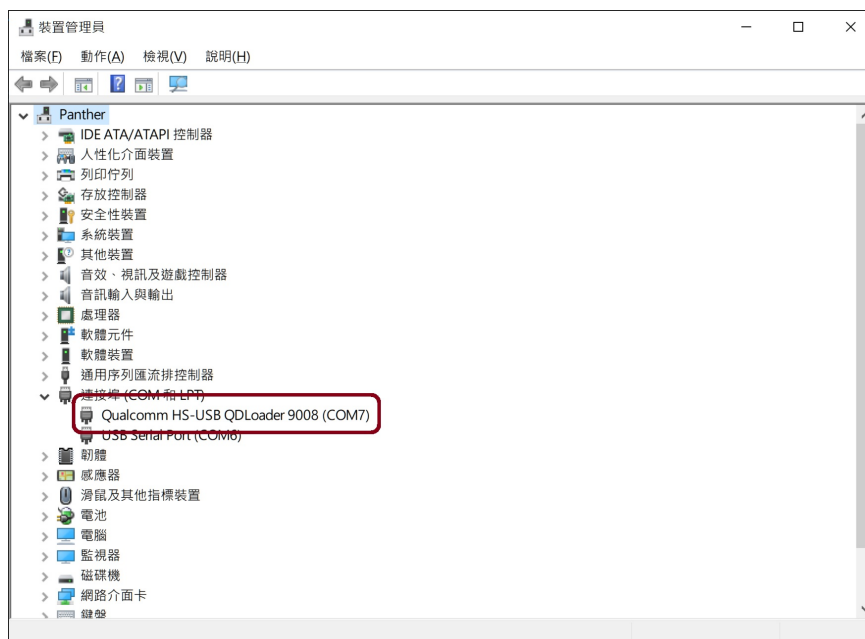
1. Force the device to enter 9008 emergency download (EDL) mode

- To enter 9008 EDL mode via force USB, follow these steps:
 - Set the Pin 1 of SW1701 for RDK or the Pin 1 of SW1500 for Grizzly switch to ON status.
 - Connect the 12 V power supply.
 - Connect the device to the host machine via USB Type-C cable
- To enter 9008 EDL mode via adb, follow these steps:

```
adb wait-for-device root
adb wait-for-device reboot edl
```

- To check if the device is under 9008 EDL mode

Open device manager in Windows and check if there is a Qualcomm HS-USB QDLoader 9008 (COMX).



Run lsusb in Linux and look for vendor ID 05c6, product ID 9008.

```
Bus 005 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 004 Device 010: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
```

2. Under 9008 EDL mode, run aim_image_loader.bat [rdk|grizzly] -1 to erase all and flash images to the device.

- RDK (aim_image_loader.bat rdk -1)
- Grizzly (aim_image_loader.bat grizzly -1)

3. Exit 9008 EDL mode, follow these steps:

- Disconnect the 12 V power supply and USB Type-C cable.
- Set the Pin 1 SW1701 for RDK or the Pin 1 of SW1500 for Grizzly switch to OFF status (opposite position to ON).
- Connect the power adaptor(12V) to the device first and then USB Type-C between the device and PC.

Method 2: Linux (qdl)

1. Run qdl_flash.sh [rdk|grizzly] script to automate the flashing process.

- RDK (qdl_flash.sh rdk)
- GRIZZLY (qdl_flash.sh grizzly)

This script will: - Check if the device is in 9008 EDL mode or 901D Mission mode.

- If the device is in 9008 EDL mode, proceed without any changes.

- If the device is in 901D Mission mode, automatically switch the device to EDL mode.

- Otherwise, check for a Qualcomm device and show an error if none is found.

Set up display on RDK

To set up the display on the device, follow these steps. 1. To enable HDMI output, pull pin 8 of SW1701 as ON. 2. Plug in HDMI cable to the external display. 3. Reboot the device. (e.g., adb reboot)

Set up cameras on RDK

Export the Wayland environment variables once per session, then run the following gst-launch-1.0 command for each camera.

```
export WAYLAND_DISPLAY=wayland-1
export XDG_RUNTIME_DIR=/run/user/root
```

1. The image for preview is available on the device display.
2. To stop the test, press the CTRL+C key combination.

For camera 0 (CON5102, IMX577)

```
gst-launch-1.0 qtiqmmfsrc name=camsrc camera=0 ! video/x-raw,format=NV12,width=1920,height=1080,framerate=30/1 ! waylandsink
fullscreen=true async=true sync=false
```

For camera 1 (CON5002, IMX586)

```
gst-launch-1.0 qtiqmmfsrc name=camsrc camera=1 ! video/x-raw,format=NV12,width=1920,height=1080,framerate=30/1 ! waylandsink
fullscreen=true async=true sync=false
```

For camera 2 (CON5001, IMX586)

```
gst-launch-1.0 qtiqmmfsrc name=camsrc camera=2 ! video/x-raw,format=NV12,width=1920,height=1080,framerate=30/1 ! waylandsink
fullscreen=true async=true sync=false
```

For camera 3 (CON5101, IMX686)

```
gst-launch-1.0 qtiqmmfsrc name=camsrc camera=3 ! video/x-raw,format=NV12,width=1920,height=1080,framerate=30/1 ! waylandsink
fullscreen=true async=true sync=false
```

For camera 4 (CON5103, IMX586)

```
gst-launch-1.0 qtiqmmfsrc name=camsrc camera=4 ! video/x-raw,format=NV12,width=1920,height=1080,framerate=30/1 ! waylandsink
fullscreen=true async=true sync=false
```

For more detail, please check https://docs.qualcomm.com/bundle/resource/topics/80-79054-300/set-up-the-camera_4_5.html#set-up-the-mipi-camera

Video playback to display with GStreamer on RDK

- H.264 video playback sample commands
 - 4K video clip: /data/test/4k.mp4
 - 1080p video clip: /data/test/1080p.mp4
 - 720p video clip: /data/test/720p.mp4

```
export XDG_RUNTIME_DIR=/run/user/root
export WAYLAND_DISPLAY=wayland-1
gst-launch-1.0 filesrc location=<path/to/mp4 file> ! qtdemux ! h264parse ! qtic2vdec ! video/x-raw(memory:GBM) ! waylandsink
fullscreen=true enable-last-sample=false
```

For more detail, please check https://docs.qualcomm.com/doc/80-79054-300/topic/verify-playback-and-preview-via-wayland-sink_4_8.html

Audio setting, playback and capture on RDK

- Check the status of the audio source and sink

- Audio source


```
pactl list sources
```
- Audio sink


```
pactl list sinks
```
- All audio ports


```
pactl list cards
```
- Switch a audio source or sink port
 - Source
 - Headset Microphone


```
pactl set-source-port regular0 headset-mic
```
 - Handset Microphone


```
pactl set-source-port regular0 handset-mic
```
 - Sink
 - Speaker


```
pactl set-sink-port 1 speaker
```
 - Headset


```
pactl set-sink-port 1 headset
```
- Audio playback with GStreamer
 - PCM


```
gst-launch-1.0 filesrc location=<path/to/wav file> ! wavparse ! audioconvert ! pulsesink volume=<0.0-1.0> 0.0 = mute, 1.0 = max volume
```
 - MP3


```
gst-launch-1.0 filesrc location=<path/to/mp3 file> ! mpegaudioparse ! mpg123audiodec ! pulsesink volume=<0.0-1.0> 0.0 = mute, 1.0 = max volume
```
- Audio playback with PulseAudio
 - PCM


```
paplay -v <path/to/wav file>
```
- Audio capture with GStreamer
 - PCM


```
gst-launch-1.0 pulserc ! audio/x-raw,format=S16LE,rate=48000,channels=2 ! audioconvert ! wavenc ! filesink location=<path/to/wav file>
```
 - MP3


```
gst-launch-1.0 pulserc ! audio/x-raw,format=S16LE,rate=48000,channels=2 ! audioconvert ! audioresample ! lameenc bitrate=192 cbr=true ! filesink location=<path/to/mp3 file>
```
- Audio capture with PulseAudio
 - PCM


```
parec -v --rate=48000 --format=s16le --channels=1 --file-format=wav <path/to/wav file>
```
- Audio capture with qtittymix
 - AMIC1 (HW: U404)
 - Set up mixer settings


```
qtittymix "TX DEC3 MUX" "SWR_MIC"
qtittymix "TX SMIC MUX3" "SWR_MIC9"
qtittymix "TX AIF1 CAP Mixer DEC3" "1"
qtittymix "ADC4 ChMap" "SWRM_TX3_CH2"
qtittymix "ADC4 MUX" "INP5"
qtittymix "TX3 MODE" "ADC_LP"
qtittymix "ADC4 MIXER Switch" "1"
qtittymix "AMIC5 MIXER Switch" "1"
qtittymix "TX_DEC3 Volume" "100"
systemctl stop pulseaudio
agmcap <path/to/wav file> -D 100 -d 101 -c 1 -r 48000 -b 16 -i CODEC_DMA-LPAIF_RXTX-TX-3
```
 - Restore initial mixer settings


```
qtittymix "TX DEC3 MUX" "MSM_DMIC"
qtittymix "TX SMIC MUX3" "ZERO"
```

```

qtitinymix "TX_AIF1_CAP Mixer DEC3" "0"
qtitinymix "ADC4_ChMap" "ZERO"
qtitinymix "ADC4_MUX" "INP5"
qtitinymix "TX3_MODE" "ADC_INVALID"
qtitinymix "ADC4_MIXER Switch" "0"
qtitinymix "AMIC5_MIXER Switch" "0"
qtitinymix "TX_DEC3_Volume" "84"

```

◦ AMIC2 (HW: U405)

■ Set up mixer settings

```

qtitinymix "TX_DEC0_MUX" "SWR_MIC"
qtitinymix "TX_SMIC_MUX0" "SWR_MIC4"
qtitinymix "TX_AIF1_CAP Mixer DEC0" "1"
qtitinymix "ADC3_ChMap" "SWRM_TX2_CH1"
qtitinymix "TX0_MODE" "ADC_LP"
qtitinymix "ADC3_MUX" "INP6"
qtitinymix "ADC3_MIXER Switch" "1"
qtitinymix "AMIC6_MIXER Switch" "1"
qtitinymix "TX_DEC0_Volume" "100"
systemctl stop pulseaudio
agmcap <path/to/wav file> -D 100 -d 101 -c 1 -r 48000 -b 16 -i CODEC_DMA-LPAIF_RXTX-TX-3

```

■ Restore initial mixer settings

```

qtitinymix "TX_DEC0_MUX" "MSM_DMIC"
qtitinymix "TX_SMIC_MUX0" "ZERO"
qtitinymix "TX_AIF1_CAP Mixer DEC0" "0"
qtitinymix "ADC3_ChMap" "ZERO"
qtitinymix "TX0_MODE" "ADC_INVALID"
qtitinymix "ADC3_MUX" "INP4"
qtitinymix "ADC3_MIXER Switch" "0"
qtitinymix "AMIC6_MIXER Switch" "0"
qtitinymix "TX_DEC0_Volume" "84"

```

◦ DMIC1 (HW: U401)

■ Set up mixer settings

```

qtitinymix "TX_DEC0_MUX" "SWR_MIC"
qtitinymix "TX_SMIC_MUX0" "SWR_MIC4"
qtitinymix "TX_AIF1_CAP Mixer DEC0" "1"
qtitinymix "DMIC1_ChMap" "SWRM_TX2_CH1"
qtitinymix "DMIC1_MIXER Switch" "1"
qtitinymix "TX_DEC0_Volume" "100"
systemctl stop pulseaudio
agmcap <path/to/wav file> -D 100 -d 101 -c 1 -r 48000 -b 16 -i CODEC_DMA-LPAIF_RXTX-TX-3

```

■ Restore initial mixer settings

```

qtitinymix "TX_DEC0_MUX" "MSM_DMIC"
qtitinymix "TX_SMIC_MUX0" "ZERO"
qtitinymix "TX_AIF1_CAP Mixer DEC0" "0"
qtitinymix "DMIC1_ChMap" "ZERO"
qtitinymix "DMIC1_MIXER Switch" "0"
qtitinymix "TX_DEC0_Volume" "84"

```

◦ DMIC2 (HW: U402)

■ Set up mixer settings

```

qtitinymix "TX_DEC0_MUX" "SWR_MIC"
qtitinymix "TX_SMIC_MUX0" "SWR_MIC4"
qtitinymix "TX_AIF1_CAP Mixer DEC0" "1"
qtitinymix "DMIC3_ChMap" "SWRM_TX2_CH1"
qtitinymix "DMIC3_MIXER Switch" "1"
qtitinymix "TX_DEC0_Volume" "100"
systemctl stop pulseaudio
agmcap <path/to/wav file> -D 100 -d 101 -c 1 -r 48000 -b 16 -i CODEC_DMA-LPAIF_RXTX-TX-3

```

■ Restore initial mixer settings

```

qtitinymix "TX_DEC0_MUX" "MSM_DMIC"
qtitinymix "TX_SMIC_MUX0" "ZERO"
qtitinymix "TX_AIF1_CAP Mixer DEC0" "0"
qtitinymix "DMIC3_ChMap" "ZERO"
qtitinymix "DMIC3_MIXER Switch" "0"
qtitinymix "TX_DEC0_Volume" "84"

```

◦ DMIC3 (HW: U403)

■ Set up mixer settings

```

qtitinymix "TX_DEC0_MUX" "SWR_MIC"
qtitinymix "TX_SMIC_MUX0" "SWR_MIC4"
qtitinymix "TX_AIF1_CAP Mixer DEC0" "1"
qtitinymix "DMIC5_ChMap" "SWRM_TX2_CH1"
qtitinymix "DMIC5_MIXER Switch" "1"
qtitinymix "TX_DEC0_Volume" "100"

```

```
systemctl stop pulseaudio
agmcap <path/to/wav file> -D 100 -d 101 -c 1 -r 48000 -b 16 -i CODEC_DMA-LPAIF_RXTX-TX-3
```

- Restore initial mixer settings

```
qtitinymix "TX DEC0 MUX" "MSM_DMIC"
qtitinymix "TX SMIC MUX0" "ZERO"
qtitinymix "TX_AIF1_CAP Mixer DEC0" "0"
qtitinymix "DMIC5 ChMap" "ZERO"
qtitinymix "DMIC5_MIXER Switch" "0"
qtitinymix "TX_DEC0 Volume" "84"
```

- Headset microphone

- Set up mixer settings

```
qtitinymix "TX DEC0 MUX" "SWR_MIC"
qtitinymix "TX SMIC MUX0" "SWR_MIC5"
qtitinymix "DEC0_BCS Switch" "1"
qtitinymix "TX_AIF1_CAP Mixer DEC0" "1"
qtitinymix "ADC2 MUX" "INP2"
qtitinymix "ADC2 ChMap" "SWRM_TX2_CH2"
qtitinymix "MBHC ChMap" "SWRM_TX1_CH3"
qtitinymix "TX1 MODE" "ADC_LP"
qtitinymix "BCS Channel" "CH2"
qtitinymix "HDR12 MUX" "NO_HDR12"
qtitinymix "ADC2_MIXER Switch" "1"
qtitinymix "AMIC2_MIXER Switch" "1"
qtitinymix "TX_DEC0 Volume" "100"
systemctl stop pulseaudio
agmcap <path/to/wav file> -D 100 -d 101 -c 1 -r 48000 -b 16 -i CODEC_DMA-LPAIF_RXTX-TX-3
```

- Restore initial mixer settings

```
qtitinymix "TX DEC0 MUX" "MSM_DMIC"
qtitinymix "TX SMIC MUX0" "ZERO"
qtitinymix "DEC0_BCS Switch" "0"
qtitinymix "TX_AIF1_CAP Mixer DEC0" "0"
qtitinymix "ADC2 MUX" "INP2"
qtitinymix "ADC2 ChMap" "ZERO"
qtitinymix "MBHC ChMap" "ZERO"
qtitinymix "TX1 MODE" "ADC_INVALID"
qtitinymix "BCS Channel" "CH0"
qtitinymix "HDR12 MUX" "NO_HDR12"
qtitinymix "ADC2_MIXER Switch" "0"
qtitinymix "AMIC2_MIXER Switch" "0"
qtitinymix "TX_DEC0 Volume" "84"
```

For more detail, please check https://docs.qualcomm.com/doc/80-79054-300/topic/set-up-audio_4_6.html

Setup AI

TensorFlow Lite using the label image app and benchmark app are used to set up AI.

Please refer to https://docs.qualcomm.com/bundle/resource/topics/80-79054-300/set-up-AI_4_10.html